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From Signals to Sensing: MatID a Deep Learning Approach to WiFi-Based Material Identification

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Abstract

This thesis explores the application of WiFi Sensing technology to interpret the surrounding environment by analyzing changes in WiFi signals. Central to this research is MatID, a novel deep learning model designed for material identification. MatID effectively combines classical techniques for material identification, such as the use of amplitude and phase, with advanced methods including spectrogram analysis. The research leverages Channel State Information (CSI) and Received Signal Strength Indicator (RSSI) to identify different materials non-intrusively. Traditional methods face challenges due to noise and signal distortions; thus, this study proposes advanced preprocessing techniques and deep learning models to enhance signal clarity and identification accuracy.

The methodology includes rigorous signal preprocessing routines such as amplitude sanitization, phase calibration, and spectrogram sanitization, which greatly improve the quality of signal data. MatID employs a combination of 2D and 1D Convolutional Neural Networks (CNNs) along with a ResNet18 to process amplitude, phase, RSSI, and spectrogram data derived from the CSI data. Experiments were conducted in both shielded and unshielded environments to simulate real-world conditions, with data collected from various materials. MatID shows significant improvement in classification accuracy, precision, and recall compared to other models tested in this study, achieving high accuracy even in the presence of noise.

This research contributes to the advancement of WiFi Sensing technology, providing a robust framework for future applications in non-invasive environmental sensing and object detection.

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Chapter 1

Introduction

In recent years, WiFi technology has become an essential part of our lives. It has enabled connectivity for devices such as laptops, phones, and smart devices. However, WiFi can have other uses beyond simple network communication. The field that studies the different uses of WiFi is called WiFi Sensing.

This technology aims to use the WiFi signal to interpret the surrounding environment, including objects, people, and places. This technology is receiving increasing interest because it represents a valid alternative to existing solutions based on visual information (Zhou, Huang, Li, Li, Li and Zhang (2019), Wang, Chen, Yang, Bi and Yu (2020)). The reasons are manifold. Firstly, WiFi allows for the maintenance of privacy since no sensitive information is captured through the WiFi signal. Moreover, WiFi technology is not affected by non-line-of-sight (NLOS) environments, darkness, smoke, or through walls. Another reason is cost. Very often, solutions based on visual information require highly sophisticated and expensive equipment, while WiFi allows for covering large areas using affordable tools that are accessible to everyone.

WiFi Sensing is used for many different purposes, such as person re-identification (Re-ID) (Avola, Cascio, Cinque, Fagioli and Petrioli (2022)) which allows to understand if a certain person is present in a place at different times, and fall detection (Wang, Yang, Li, Wu and Wang (2021)) which allows to understand if a person has fallen. This technology is very useful for the elderly, as it allows for promptly requesting help.

WiFi is a technology that uses radio signals transmitted from different access points (APs) to enable communication between different devices. The 802.11bf Wi-Fi standard offers a set of protocols that facilitate the establishment of Wi-Fi stations, forming the basis for a WiFi Sensing network. When the WiFi signal travels between two connected devices, it encounters objects or people on its path. Once these are crossed, however, the WiFi signal changes (Chen, Zhang, Jiang, Cao and Cui (2019), Bialer, Raphaeli and Weiss (2020)).

These changes can be measured through the received signal strength indicator (RSSI) or the channel state information (CSI). The RSSI is a single value computed at the MAC layer that represents the signal quality. Instead, CSI is measured by employing the orthogonal frequency-division multiplexing (OFDM) transmission technology at the PHY layer and includes fine-grained signal information defined at the subcarrier level (Yang, Zhou and Liu (2013)). Between these two measurements,

CSI has been shown to be more robust and carries more information than RSSI. Most WiFi sensing applications use amplitude and phase values that can be extracted from the CSI (Li, Qu, Liu, Zhou, Sun, Sigg and Li (2021), Rao, Li, Yang and Wang (2020), Xiao, Chen, Xie, Gui, Sun and Wang (2020)).

However, the WiFi signal also has negative aspects related to noise. For this reason, most sensing applications need to apply preprocessing on the data to eliminate noise and improve the information contained in the CSI. The most used techniques consist of removing the amplitude outliers (Dang, Si, Hao and Huang (2019)) and the phase offset (Wang, Gao and Mao (2015)). Afterward, an approach based on neural networks is used to solve the task. WiFi can also be used for material identification. Material identification has a broad range of applications, notably in the field of security. An early instance of this technology is Wi-Metal (Wu (2016)), designed to detect potential metallic objects using WiFi. This technology was subsequently enhanced to enable in-baggage object identification (Shi et al. (2021)), to detect potentially suspicious items within luggage using cost-effective WiFi devices. Furthermore, material identification is employed to differentiate between metallic and non-metallic objects. Several studies have utilized WiFi to identify whether a substance passed through is liquid or solid (Gu et al. (2021)).

1.1 State of the Art

Material identification has great potential. However, it is currently a field that has not been extensively explored.

Among the most notable studies, we find WiMi (Feng, Xiong, Chang, Wang, Chen, Fang and Tang (2019)), a non-contact target material identification system, implemented on ubiquitous and cheap commercial off-the-shelf (COTS) Wi-Fi devices. The intuition is that different materials produce different amounts of phase and amplitude changes when a target appears on the line-of-sight (LoS) of a radio frequency (RF) link. In particular, information regarding phase and amplitude can be derived from the channel state information (CSI). Additionally, the phase change is defined as the difference between two phase readings, which are measured before and after a target appears at the LoS link, while the amplitude change is defined as the difference between the two amplitude readings measured before and after a target appears at the LoS link. However, the raw phase and amplitude extracted are unusable as they are very noisy. Therefore, signal sanitization is necessary. In particular, WiMi implements phase calibration, to address phase errors caused by asynchronization between transceivers and hardware noise, the technique uses a second antenna, which is common in Wi-Fi devices. This helps to remove noise differences and stabilize phase readings. By comparing the phase readings from two closely-placed antennas, the method achieves more stable phase difference values, which are crucial for accurate material identification. The technique involves selecting 'good' subcarriers with smaller phase difference variances, which are less affected by multipath, to further improve stability and accuracy. WiMi also performs amplitude denoising, which consists of two steps. First, outlier removal is performed, a technique that calculates the mean value μA and standard deviation σA of the measured CSI amplitudes. After data samples whose amplitude values fall within

the range [μA - 3σA, μA + 3σA] are preserved while all other values are discarded. After this, the authors observed that despite outlier removal, there was still a significant amount of impulse noise. Therefore, to remove it, they observed that wavelet coefficients of the useful signals are often strongly correlated in different scales (frequencies), while the wavelet coefficients of the noises are weakly correlated or uncorrelated. Then by multiplying the wavelet coefficients of adjacent scales, the original useful signals can be reconstructed since they will be larger than the noise. WiMi successfully identifies 10 commonly seen liquids at an overall accuracy higher than 95% with strong multipath indoors.
Shi et al. (2021) proposed an environment-independent in-baggage object identification system by leveraging low-cost WiFi. In this work, WiFi features based on polarized directional antennas are used to capture objects' material and shape characteristics. Then, a convolutional neural network-based model is employed to integrate the WiFi features and perform accurate in-baggage object identification. Additionally, material-based domain adaptation using adversarial learning is presented to facilitate fast deployments in different environments.

Polarized WiFi signals usually oscillate on a single plane (or two planes for dual-polarization), while unpolarized WiFi signals could oscillate in all planes perpendicular to the propagation direction. The polarized WiFi signal can be decomposed into two orthogonal polarized directions, including p-polarization, with the oscillation direction on the plane of incident, and s-polarization, with the oscillation direction perpendicular to the plane of incident. Upon bouncing off the object's surface, the polarization of WiFi signals can carry information reflecting the material of the object. One of the key ideas of this paper is that if polarized WiFi hits a non-metallic object, the reflected signal will always be polarized and will contain both s- and p-polarization components. Conversely, a metallic object will depolarize the reflected WiFi signal, and the electromagnetic wave can vibrate in all directions perpendicular to the propagation direction. By measuring the polarization of the reflected WiFi signal, it is possible to identify the material. In particular, the study was conducted by collecting the CSI (Channel State Information) of the reflected WiFi signal from four cylinders of the same shape and size but with different materials. However, although WiFi polarization can be used to distinguish between metals and non-metals, it does not capture shape characteristics. For this reason, the paper introduces the CSI complex difference, which uses the complex difference of CSI between two receiving antennas to encode information related to the position and characteristics of objects. Feature extraction is a fundamental part of this work. First, signal cleaning of CSI is performed, particularly since the CSI phase can be distorted by the sampling time offset (STO) and sampling frequency offset (SFO) between the transmitter and the receiver. To calibrate the phase, CSI magnitude and phase are calculated for all subcarriers. Then, a linear transformation-based method is used to correct phase errors in WiFi signals caused by temporal and frequency misalignments. Subsequently, a moving average filter is applied to the time-series magnitude and phase values to remove such noises. Finally, given the magnitude and phase of denoised CSI, the complex form of CSI is reconstructed and used to derive CSI complex difference and polarization features. To extract polarization features, the ratio of signal power between two orthogonal polarized antennas is measured, which characterizes the oscillation direction of the electromagnetic wave

while removing amplitude attenuation due to signal propagation. The CSI complex difference is obtained from two receiving antennas using the following formula:

$$\hat{H}_i = H_i^{t\leftrightarrow r_1}(H_i^{t\leftrightarrow r_2})^* = |\hat{H}_i|e^{j\angle\hat{H}_i} \tag{1.1}$$

where $H_i^{t\leftrightarrow r_1}$ and $H_i^{t\leftrightarrow r_2}$ denote the CSI from the links $t \leftrightarrow r_1$ and $t \leftrightarrow r_2$, respectively. $(.)^*$ denotes the complex conjugate.

Based on the magnitude and angle of the CSI complex difference, seven statistical features are calculated, including mean, variance, median, skewness, kurtosis, IQR, and range. The system architecture is shown in Figure 1.1.

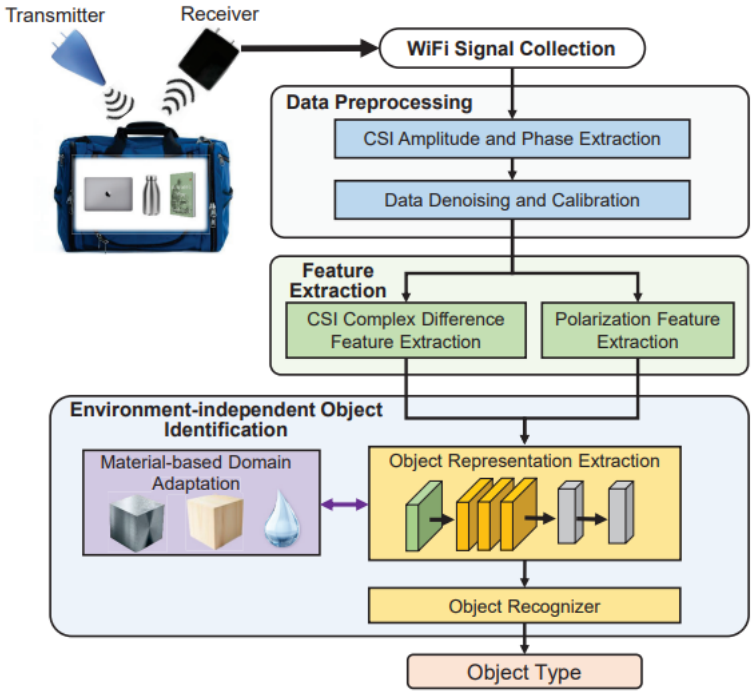


Figure 1.1 System architecture of the proposed method Shi et al. (2021).

The model presented in this work consists of a CNN that serves as an Object Representation Extractor, whose task is to map the CSI complex difference and polarization features into a set of object representations. Then, an Object Recognizer is developed to identify objects based on the derived object representation. To mitigate domain impacts and enable environment-independent in-baggage object identification, a Domain Adaptation Framework using adversarial training is created. The input of the CNN model has a size of (56,8) where 56 represents the number of subcarriers while 8 are the extracted features, seven CSI complex difference statistics, and one polarization feature from each of the 56 subcarriers. A batch normalization layer is used to perform normalization, which reduces small-scale sample inconsistency by removing the mean and scaling the samples to unit variance. Three convolutional layers are then used to perform 2D convolution and calculate the 2D representations, which integrate all input features constructively. A dropout layer is attached to each convolutional layer to prevent over-fitting. The 2D representations from the third convolution layer are flattened and compressed by a fully connected

layer with 128 neurons. By taking the input features X, the CNN model maps the input into object representations Z as follows:

$$Z = CNN(X, \theta) \tag{1.2}$$

where θ represents a set of learnable parameters in the representation extractor. The activations of convolutional layers and the fully connected layer are all Leaky ReLU. Based on the extracted object representation Z, a neural network with two fully connected layers is then used to perform object identification. The number of neurons for the two layers is 256, and the activations are Leaky ReLU. Finally, a SoftMax layer is used to predict the type of object. Additionally, a domain discriminator is created to transfer the material characteristics as a whole from the source domain to the target domain. This domain discriminator has the same structure as the object recognizer described previously. The flow of the domain adaptation framework is shown in Figure 1.2. Using this method, the authors were able to achieve exceptional results, achieving an accuracy of over 97% in identifying in-baggage objects in the same environment, demonstrating how polarization characteristics and the complex difference of Channel State Information (CSI) effectively capture material and shape characteristics of objects. Furthermore, thanks to their domain adaptation method, the accuracy of object identification improved by 42% when the system was deployed in a new environment with little training.

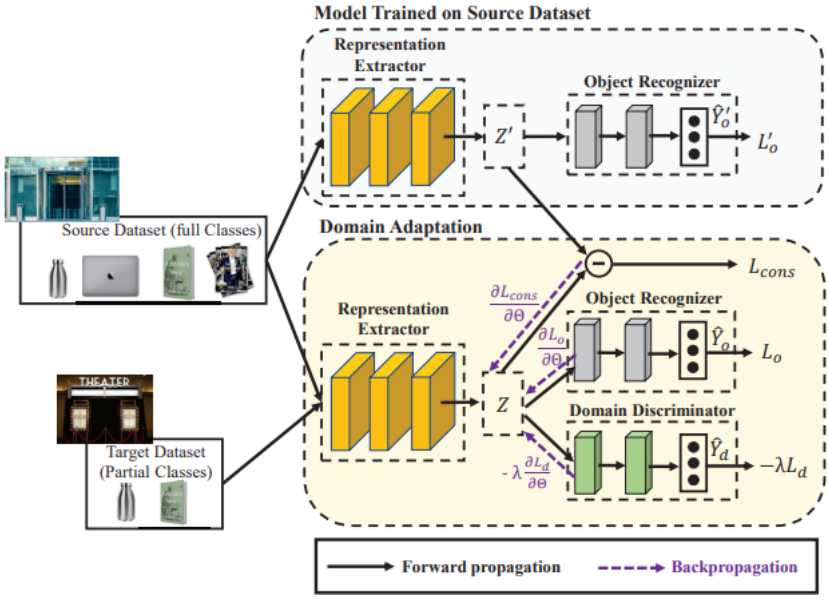


Figure 1.2 Flow of the proposed domain adaptation framework. Shi et al. (2021).

Finally, we have WiMate (Gu et al. (2021)), a non-contact material identification system leveraging only off-the-shelf WiFi devices. The key enabler of WiMate is a novel theoretical model used to characterize how the electromagnetic wave decays when penetrating different materials. The proposed model identifies a unique feature for each material that only depends on the material itself. The underlying idea of this method is that when the microwave propagates through a dissipative medium,

its amplitude is attenuated by $e^{-K_I \cdot D}$. Meanwhile, the medium's phase transition is measured as $K_R \cdot D$, where D is the propagation distance, which is equal to the material thickness. K_I and K_R can be calculated as follows:

$$K_i = \omega\sqrt{\mu_0\epsilon_0\epsilon_r}\left[\frac{1}{2}\left(\sqrt{1+\frac{\sigma^2}{(\epsilon_0\epsilon_r\omega)^2}}-1\right)\right]^{\frac{1}{2}} \tag{1.3}$$

$$K_r = \omega\sqrt{\mu_0\epsilon_0\epsilon_r}\left[\frac{1}{2}\left(\sqrt{1+\frac{\sigma^2}{(\epsilon_0\epsilon_r\omega)^2}}+1\right)\right]^{\frac{1}{2}} \tag{1.4}$$

where ω is the angular frequency of the wave, μ_0 and ϵ_0 are the magnetic permeability and the dielectric constant under vacuum, respectively. ϵ_r and σ are the relative dielectric constant of the medium and the electrical conductivity of the medium. K_I and K_R are solely dependent on the material. When placing the material to be identified on the line-of-sight (LOS) path between the transmitting antenna (T_X) and the receiving antenna (R_X), the channel frequency response can be rewritten as:

$$h_{R_1}(f,t)\cdot e^{j\theta_{R_1}} = h'_l(f,t)\cdot e^{-K_1\cdot D}\cdot e^{j(\theta'_l - K_r\cdot D)} + h_m(f,t)\cdot e^{j\theta_m} \tag{1.5}$$

where $h_{R_1}(f,t)\cdot e^{j\theta_{R_1}}$ represents the complex value channel frequency responses in the format of CSI, $h'_l(f,t)$ and θ' represent signals that are not affected by the material.

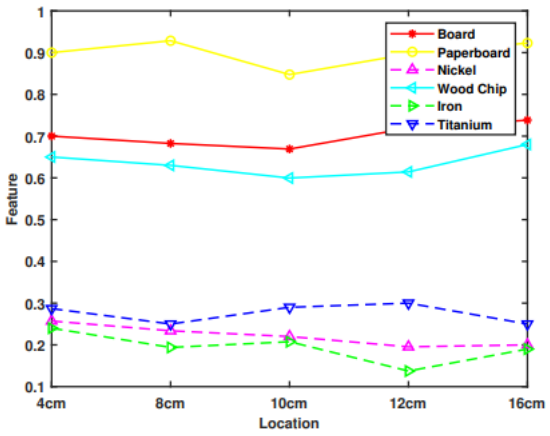
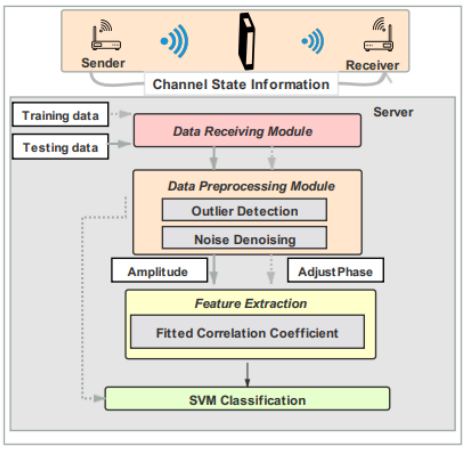


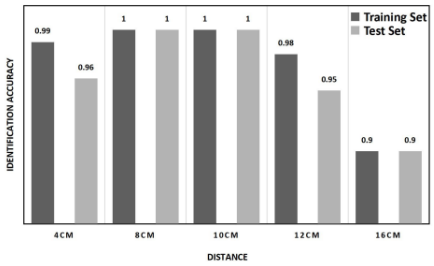
Figure 1.3 Feature value of six different materials at different locations Gu et al. (2021).

The preprocessing phase is very similar to that of previous methods, where the CSI is extracted, and from this, amplitude and phase are derived. Subsequently, to make them more meaningful, amplitude and phase are sanitized, particularly using the outlier removal method. Next, to further reduce noise, a Butterworth filter is applied. Once the signal sanitization is completed, feature extraction is performed. Amplitude and phase are divided into groups of 1000 data packets, and the average value within the group is calculated as the amplitude and phase difference value of the group. Then, the theoretical model is used to calculate the

feature value of the group. As shown in Figure 1.3, the features successfully identify different materials in different environments. For classification, an SVM is used. The complete architecture of the system is shown in Figure 1.4a.



(a) WiMate complete architecture Gu et al. (2021).



(b) Classification accuracy of the model in different locations Gu et al. (2021).

This model achieves an average accuracy of 96.7% for 6 different materials in 5 different locations (Figure 1.4b).

1.2 Proposed Solution

This study introduces MatID, a novel deep learning model for material identification. The primary feature of MatID lies in its effective combination of classical techniques for material identification, such as the use of amplitude and phase, and the use of the spectrogram, widely employed in WiFi-based fall detection (Nakamura, Bouazizi, Yamamoto and Ohtsuki (2020)). Specifically, once amplitude and phase are extracted from the CSI signal, they are respectively sanitized through outliers removal and calibration. The spectrogram is then derived from the CSI signal; however, it requires a denoising process to make it usable. For this purpose, the Hann window is employed because, due to its characteristic of attenuating the left and right sides of the window, it ensures minimal differences at the extremes. This allows not only noise reduction but also prevents spectral loss. Additionally, the received signal strength indicator (RSSI) is also utilized. This metric represents the signal power, indicating the signal strength expressed in dBm. After this preprocessing work, the sanitized data are fed into the network. The network comprises two CNNs, one 2D and another 1D. Amplitude, phase, and RSSI are passed to the 2D CNN. The 1D CNN, on the other hand, is tasked with processing the phase. The spectrogram, meanwhile, is passed to a ResNet18 to extract as much information as possible. The dataset used to train the model includes measurements obtained in a shielded environment, i.e., in a shielded box without external interferences, as well as measurements in an unshielded environment, i.e., with external interferences without a shielded box.

1.3 Thesis Outline

The thesis is structured into five chapters, beginning with an introduction to provide context and set the stage for the subsequent discussions. Chapter Two delves into the intricacies of Wi-Fi, exploring its functionalities, characteristics, and various applications. It also examines signal sanitization techniques crucial for the project's context. In Chapter Three, the methodology is outlined, focusing on the two primary networks utilized: Convolutional Neural Network (CNN) and Residual Neural Network (ResNet). This chapter elucidates the rationale behind their selection and their contributions to the research objectives. Chapter Four provides a comprehensive summary of the experimental results, including the outcomes of CNN and ResNet training and the accuracy in signal identification. Additionally, it outlines the iterative process that led to the development of the final architecture and compares the results obtained from the implemented networks. Finally, Chapter Five offers a conclusive wrap-up of the thesis, presenting overarching conclusions drawn from the research findings. It succinctly discusses the significance of these findings and explores potential avenues for future enhancements.

Chapter 2

The Wireless Channel

2.1 Fundamentals of Wireless Communication

All wireless technologies, Wi-Fi included, rely on electromagnetic (EM) waves to transfer information between a transmitting (TX) and a receiving (RX) antenna. These waves move at a consistent pace, manifesting as sinusoidal patterns when traversing through mediums like air or space. EM waves, birthed from electric and magnetic field oscillations, carry distinct wavelength and frequency attributes. The wavelength is the distance between successive wave peaks or valleys, quantified in meters (m). Frequency, instead, represents oscillation cycles per second, measured in Hertz (Hz).

A crucial relationship exists between wave speed v, wavelength λ, and frequency f, described by:

$$v = \lambda f \tag{2.1}$$

Here, v denotes the speed of light in a vacuum, around 3×10^8 m/s, or a different value in other mediums. From this formula, we derive:

$$\lambda = \frac{v}{f} \tag{2.2}$$

$$f = \frac{v}{\lambda} \tag{2.3}$$

This relationship emphasizes that frequency and wavelength are directly proportional to the wave's speed and inversely related to each other.

The EM field produced by traveling radiation alters based on its distance from the source, as depicted in Figure 2.1. The area within a single wavelength of the TX antenna is called the near-field. Within this zone, radiation strength diminishes swiftly as it moves away from the source, with nearby receivers affecting the transmitter's radiation. This near-field plays an important role in NFC communication, necessitating the receiver NFC tag to be within 4 cm of the transmitter. Beyond this lies the far-field, starting two wavelengths from the TX antenna, characterized by power reduction inversely proportional to the square of the distance from the TX antenna. In this region, the original wave radiated by the TX antenna remains unaffected by receivers, facilitating long-range transmissions crucial for most wireless

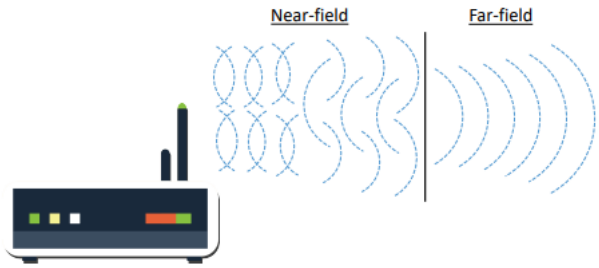


Figure 2.1 Electromagnetic field regions: near-field and far-field.

sensing applications using prevalent radio communication methods like Bluetooth or Wi-Fi.

Defining a physical model for a wireless channel Costello (2009) begins with a stationary TX antenna emitting waves into space. Given the proportional and perpendicular nature of electric and magnetic fields concerning each other and the wave's propagation direction, it's feasible to focus on one field to describe the transmitting source's far-field. When a sinusoidal wave $\cos(2\pi ft)$ is transmitted in a vacuum without an RX antenna, the electric far-field at time t is modeled as:

$$E(f,t,(r,\theta,\psi)) = \text{`}\frac{\alpha_s(\theta,\psi,f)\cos 2\pi f(t-\frac{r}{c}))}{r} \tag{2.4}$$

Here, r signifies the distance from the transmitter, θ and ψ represent horizontal and vertical angles from the TX antenna, c is the speed of light, and $\alpha_s(\theta,\psi,f)$ stands for the transmitting device's far-field radiation pattern at frequency f.

Antenna radiation patterns depict the power distribution across angles of the radiated EM wave. These patterns comprise radiation lobes, with the main lobe being the direction of peak radiation power. Conversely, side lobes and back lobes denote less desirable radiation directions with reduced power. Electromagnetic reciprocity dictates that an antenna's receiving pattern matches its far-field radiation pattern when utilized for transmission.

Given a fixed RX antenna at a specific position $x = (r,\theta,\psi)$ and assuming no noise, the waveform received from a sinusoidal transmission is linearly represented as:

$$E_r(f,t,x) = \frac{\alpha(\theta,\psi,f)\cos 2\pi f(t-\frac{r}{c})}{r} \tag{2.5}$$

Here, $\alpha(\theta,\psi,f)$ represents the combined transmitting and receiving patterns of the TX and RX antennas in the direction (θ,ψ). Incorporating the RX antenna alters the electric field's behavior near x, factored in by its receiving pattern. Given the linearity of Equation 2.4 and Equation 2.5, the EM field response at x to a weighted transmitted waveform sum is the weighted sum of each waveform's responses. Consequently, for a location x, we can formulate:

$$H(f) := \frac{\alpha(\theta,\psi,f)e^{-j2\pi f\frac{r}{c}}}{r} \tag{2.6}$$

Resulting in $E_r(f,t,x) = \mathcal{R}[H(f)e^{j2\pi ft}]$. Thus, $H(f)$ corresponds to Equation 2.5 and defines the system function of a linear time-invariant channel in the frequency domain, with its inverse Fourier transform being the corresponding time-domain impulse response. However, this time-invariant characteristic is compromised if the antennas or potential obstructions are in motion relative to each other.

Considering the stationary TX antenna and the free space model outlined by Equation 2.4, for a point moving at velocity v in the direction of increasing r from the wave transmitter, described by $r(t) = r_0 + vt$, the resulting electric far-field at time t is:

$$E(f,t,(r_0+vt,\theta,\psi)) = \frac{\alpha_s(\theta,\psi,f)\cos 2\pi f(t-\frac{r_0}{c}-\frac{vt}{c})}{r_0+vt} \tag{2.7}$$

Here, $f(t-\frac{r_0}{c}-\frac{vt}{c})$ can be reformulated as $f(1-\frac{v}{c})t-\frac{fr_0}{c}$, focusing on the Doppler shift Giordano (2009) of $-\frac{fv}{c}$ induced by the point's motion away from the transmitter. This implies the EM wave's frequency f shifts to $f(1-\frac{v}{c})$. As a consequence, each successive crest in the transmitted wave advances slightly further before being detected at the observed point. Subsequently, adhering to the model defined by Equation 2.5, if the RX antenna is positioned at a mobile location $x(t) = (r(t),\theta,\psi)$, with $r(t) = r_0 + vt$, the received waveform is:

$$E(f,t,(r_0+vt,\theta,\psi)) = \frac{\alpha_s(\theta,\psi,f)\cos 2\pi f[(1-\frac{v}{c})t-\frac{r_0}{c}]}{r_0+vt} \tag{2.8}$$

The Equation 2.8 cannot be depicted as a linear time-invariant channel system function in its current form. Nonetheless, by disregarding the time-dependent distance $r(t)$ in the denominator, the channel can still be characterized as a system function, considering the Doppler shift $-\frac{fv}{c}$ instead of the original frequency f.

2.2 Linear System for Wireless Channel

In the preceding section, we characterized the wireless channel in response to a sinusoidal input $\phi(t) = \cos(2\pi ft)$ propagating into free space, considering both stationary and mobile receiving antennas. However, real-world scenarios necessitate accounting for the propagation delay of radio waves due to the distance covered at the medium's propagation speed. Additionally, during signal propagation, various attenuation factors lead to fluctuations in wireless channel strength over time and frequency, attributed to large- and small-scale fading Costello (2009).

Large-scale fading, generally frequency-independent, stems from signal path loss relative to distance and shadowing effects by substantial obstructions, like buildings, along the propagation route. Conversely, small-scale fading, frequency-dependent, arises from constructive and destructive interference due to multi-path effects, causing multiple signal replicas reaching the receiver across various paths to interfere in amplitude and phase. Thus, a multi-path wireless channel at frequency f can be linearly represented as:

$$y(f,t) = \sum_i a_i(f,t)\phi(t-\tau_i(f,t)) \tag{2.9}$$

Here, $y(f,t)$ denotes the received signal in response to a transmitted sinusoid $\phi(t)$, $a_i(f,t)$ and $\tau_i(f,t)$ represent signal attenuation and propagation delay, respectively, at time t along the i-th path between TX and RX antennas. Assuming individual path attenuation and propagation delay remain frequency-independent, the principle of superposition simplifies Equation 2.9 to:

$$y(t) = \sum_i a_i(t)x(t - \tau_i(t)) \tag{2.10}$$

Here, $x(t)$ signifies an arbitrarily transmitted bandpass signal. Although individual paths were assumed to be frequency-independent, the channel response can vary with frequency due to multi-path fading. Since Equation 2.10 is linear, a fading multi-path wireless channel can be modeled in the time domain as the Channel Impulse Response (CIR) at time t of a linear time-variant channel:

$$h(\tau,t) = \sum_i a_i(t)\delta(\tau - \tau_i(t)) \tag{2.11}$$

In Equation 2.9 and Equation 2.11, $a_i(t)$ and $\tau_i(t)$ denote the attenuation factor and propagation delay of the i-th path, respectively, while $\delta(\tau)$ corresponds to the Dirac delta function. Consequently, the relationship between the received signal at time t and the bandpass signal x transmitted at $t - \tau$ can be articulated in terms of the impulse response as:

$$y(t) = \int_{-\infty}^{\infty} h(\tau,t)x(t - \tau)d\tau \tag{2.12}$$

Applying the Fourier Transform (FT) to the impulse response allows estimation of the time-varying Channel Frequency Response (CFR):

$$H(f;t) = \int_{-\infty}^{\infty} h(\tau,t)e^{-j2\pi f\tau}d\tau = \sum_i a_i(t)e^{-j2\pi f\tau_i(t)} = |H(f;t)|e^{j\angle H(f;t)} \tag{2.13}$$

Here, $|H(f;t)|$ and $\angle H(f;t)$ represent signal amplitude and phase responses, respectively, with j being the imaginary component. In the frequency domain, a time-variant channel is linearly modeled as:

$$y(t) = H(f;t)x(t - \tau) \tag{2.14}$$

In a static environment with fixed TX and RX antennas, the wireless channel can be described as a Linear Time-Invariant (LTI) system, with Equation 2.11 and Equation 2.14 specified as:

$$h(\tau) = \sum_i a_i\delta(\tau - \tau_i) \tag{2.15}$$

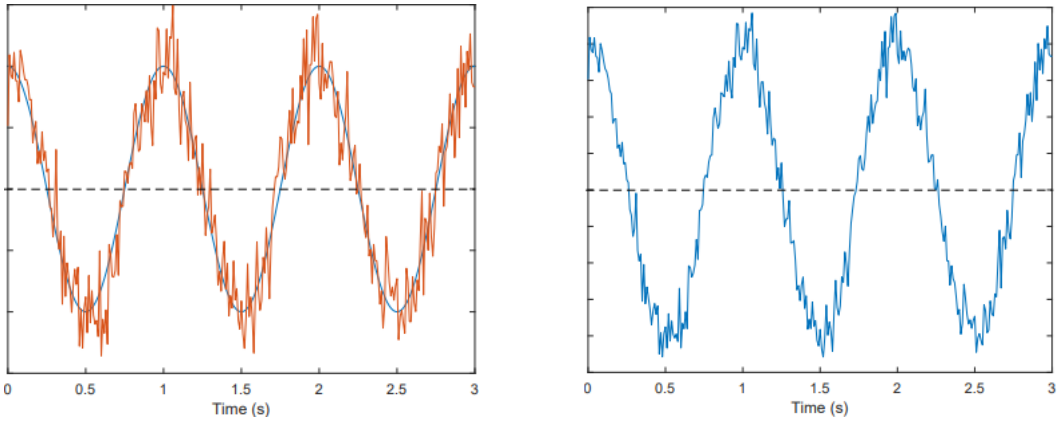
$$H(f) = \int_{-\infty}^{\infty} h(\tau)e^{-j2\pi f\tau}d\tau = \sum_i a_i e^{-j2\pi f\tau_i} = |H(f)|e^{j\angle H(f)} \tag{2.16}$$

Here, attenuations a_i and propagation delays τ_i remain constant over time t. Finally, in defining the wireless communication channel as a linear system, factors

causing signal attenuation and propagation delay must be considered alongside factors modeled as random noise. Such noise, universally encountered in real-world communication systems, is depicted in Figure 2.2, as a completely noise-free channel is physically unattainable. Additive White Gaussian Noise (AWGN) typically models the received signal noise:

$$y(t) = \sum_i a_i(t)x(t - \tau_i(t)) + \omega(t) \tag{2.17}$$

Here, $\omega(t)$ represents the AWGN component at time t. This noise component is additive, as the RX antenna processes the sum of a noise-free component, i.e., the signal received in the absence of noise, and the noisy component, independent of the transmitted signal. This component is random and drawn from a zero-mean Gaussian distribution, with Gaussian chosen due to the addictive nature of multiple independent factors, allowing the application of the Central Limit Theorem Fischer (2011), which posits that the sum of independent random variables approaches a normal distribution. In conclusion, the term "white noise" signifies uniform power distribution across the entire frequency spectrum.



(a) **(b)**

Figure 2.2 A typical example of noisy signal. In (a) the noise-free component is indicated in blue and AWGN component in orange. In (b) the signal affected by AWGN noise.

2.3 The Received Signal Strength Indicator (RSSI)

The signal received at an RX antenna undergoes a decrease in waveform power intensity as electromagnetic (EM) radiation moves away from the transmitter, interacting with various obstacles in its path. The Received Signal Strength Indicator (RSSI) serves as a metric indicating the relative signal quality, capturing fluctuations in wireless channel strength. This measurement has been proven effective for evaluating link quality in wireless networks Vlavianos, Law, Broustis, Krishnamurthy and Faloutsos (2008) and for developing various Wi-Fi sensing applications Booranawong, Jindapetch and Saito (2018), Hoang, Yuen, Dong, Lu, Westendorp and Reddy (2019). Formally, RSSI is quantified in decibels (dB) and is typically represented using the

log-distance path loss model Assayag, Oliveira, Souto, Barreto and Pazzi (2021), expressed as:

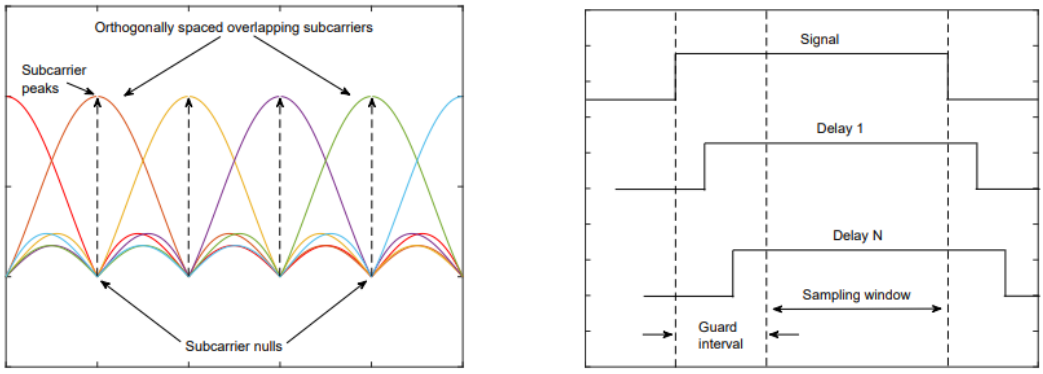
$$PL_d = PL_{d0} + 10n \log_{10}\left(\frac{d}{d_0}\right) + \chi \tag{2.18}$$

Here, PL_d denotes the path loss at a given distance d, n is the path loss exponent influenced by the environment, and χ is a zero-mean Gaussian-distributed random variable with a standard deviation σ in dB, which models noise to account for shadowing variations. Finally, PL_{d0} signifies the path loss at a reference distance d_0, typically calculated using the free-space path loss model derived from the Friis transmission equation Friis (1946), given by:

$$FSP_L = 10 \log_{10}\left(\frac{4\pi df}{c}\right)^2 \tag{2.19}$$

In Equation 2.19, d represents the distance between the TX and RX antennas, c is the speed of light, and f denotes the radio wave frequency, calculated by dividing the speed of light by the wavelength.

2.4 Orthogonal Frequency-Division Multiplexing



(a) (b)

Figure 2.3 Simplification of OFDM technology through graphical representations. In (a) is represented an OFM signal modulation example, in (b) the guard visual interpretation.

Most contemporary wireless communication systems, utilize Orthogonal Frequency-Division Multiplexing (OFDM) for signal modulation Chang (1966), Weinstein (2009). OFDM enables the transmission of data encoded across multiple orthogonal carrier frequencies, known as subcarriers, within a single communication channel. As illustrated in Figure 2.3a, a high-rate data stream is transmitted via closely spaced orthogonal subcarriers, each carrying information in parallel. These subcarriers are modulated using a digital modulation scheme with a low symbol rate.

When applying a modulation scheme to a carrier, sidebands are produced, causing interference between different subcarriers in a multi-carrier scenario due to overlapping frequencies. However, this interference is mitigated with orthogonal

frequencies, as the orthogonality property generates signal nulls between adjacent subcarriers, aligning each subcarrier peak accordingly. Consequently, the receiver can retrieve the original signal by correlating it with a known set of orthogonal sinusoids, eliminating interference from overlapping sidebands.

Furthermore, by combining multiple carrier frequencies, it becomes feasible to achieve data rates comparable to single-carrier modulation techniques within the same bandwidth. This is achieved while averting interference or signal degradation caused by the multi-path effect. This mitigation is facilitated by using guard intervals, ensuring that delayed replicas of the received signal do not disrupt its timing and phase, as depicted in Figure 2.3b.

2.5 Channel State Information

Channel State Information (CSI) is a more detailed and nuanced metric for wireless communication channel assessment compared to the previously discussed RSSI. In modern wireless communication systems based on the OFDM framework, CSI provides comprehensive insights into signal propagation characteristics from the transmitter to the receiver at the subcarrier level. This capability allows for adaptive transmission strategies tailored to current link conditions, facilitating reliable communications in Multiple-Input Multiple-Output (MIMO) antenna configurations, as depicted in Figure 2.4.

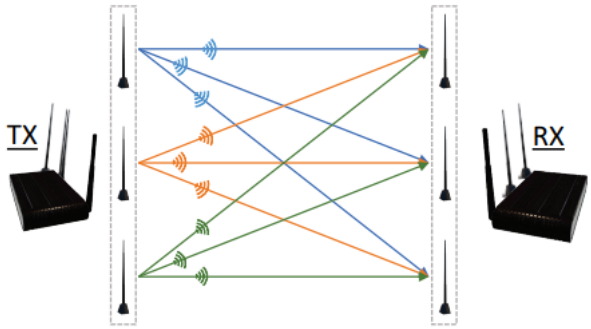


Figure 2.4 MIMO antenna configuration. Multiple TX antennas transmit multiple parallel signals at the same frequencies to multiple RX antennas.

While RSSI offers a relative measure of received signal quality, CSI provides richer information about the signal, encompassing attributes such as amplitude, phase, and frequency. Consequently, CSI has gained increasing prominence in recent years for developing a myriad of Wi-Fi sensing applications, including human presence detection Gong, Yang, Zhou, Man, Cai, Zhou and Yang (2016), activity recognition Wang, Liu, Shahzad, Ling and Lu (2017), and tracking Qian, Wu, Yang, Liu and Jamieson (2017).

In a standard Wi-Fi transmission scenario, P data packets characterize the signal exchanged between Transmitting (TX) and Receiving (RX) Access Points (APs). CSI is a frequency-domain evaluation metric that involves computing Channel Frequency Response (CFR) values for all K OFDM-based subcarriers associated

with each packet $p \in P$ reaching the receiver.

Given arrays Θ and Γ representing fixed receiving and transmitting antennas in a static environment, respectively, the frequency response $H(f)_{\theta,\gamma,k}$ for each subcarrier $k \in K$ in the wireless communication established between the $\theta \in \Theta$ and $\gamma \in \Gamma$ antennas can be expressed as:

$$H(f)_k^{\theta,\gamma} = |H(f)_k^{\theta,\gamma}| e^{j\angle H(f)_k^{\theta,\gamma}} \tag{2.20}$$

where $|H(f)_{\theta,\gamma,k}|$ denotes the signal amplitude, $\angle H(f)_{\theta,\gamma,k}$ represents the signal phase, and j is the imaginary component resulting from the Fourier transform applied to the impulse response.

Consequently, the final CSI measurement estimated across all K subcarriers, considering all antennas in the TX and RX arrays, is represented as a complex matrix of size $\Theta \times \Gamma \times K$, defined as:

$$\text{CSI} = \begin{bmatrix} H(f)_1^{(1,1)} & \dots & H(f)_k(1,1) \\ \vdots & \ddots & \vdots \\ H(f)_1^{(\theta,\gamma)} & \dots & H(f)_{(\theta,\gamma),k} \end{bmatrix} \tag{2.21}$$

where $H(f)_k^{(\theta,\gamma)}$ represents an 8-bit signed complex number indicating the k-th subcarrier CFR value over the $\theta \in \Theta$ and $\gamma \in \Gamma$ antennas.

2.6 Amplitude

One of the primary signal characteristics extracted from the Channel State Information (CSI) measurement is its amplitude. As radio waves propagate through the transmission medium, they induce patterns of disturbance that oscillate across the electromagnetic (EM) spectrum. Theoretically, the amplitude of such a measurement represents the maximum displacement of points on the EM wave from its equilibrium position (Figure 2.5).

In practical terms, however, the amplitude is often expressed as the power of the radio signal, typically reported on a decibel (dB) scale. In real-world scenarios, wireless channels are rarely noise-free communication systems. Consequently, they are appropriately modeled as $y(t) = \sum_i a_i(t)x(t - \tau_i(t)) + \omega(t)$.

It's worth noting that while amplitude patterns can be extracted from the CSI matrix, they often necessitate additional processing to mitigate potential noise stemming from wireless protocol specifications and ambient conditions. One viable solution to this challenge involves employing outlier removal strategies Dang et al. (2019).

2.6.1 Amplitude Sanitization

Abnormal values can occasionally appear in Channel State Information (CSI) measurements, potentially skewing the signal amplitude extraction procedure. To mitigate this issue, outlier removal is essential. Employing a filtering-based strategy can effectively sanitize the amplitude, eliminating irrelevant radio information that may not be essential for the intended Wi-Fi sensing task.

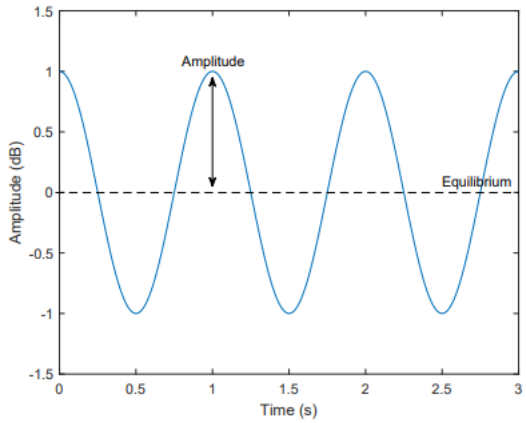


Figure 2.5 Typical example of EM wave amplitude.

To identify outlier values, the Hampel identifier Davies and Gather (1993) can be employed, defining an outlier as any point falling outside a closed interval $[\mu - \xi\sigma, \mu + \xi\sigma]$, where μ and σ represent the median and Median Absolute Deviation (MAD) of the data sequence, respectively. Here, ξ is an application-dependent constant. Specifically, for a sliding window of fixed length w, outliers are detected based on points deviating more than ξ local MAD away from the local median within this window across wireless data packets for each subcarrier.

First, local outliers are identified by leveraging local median values over the sliding window. Subsequently, the detected abnormal values are replaced with the previous non-outlier number to maintain consistent amplitude information. Formally, the local median $\mu(\Omega^{p,k})$ is defined as:

$$\mu(\Omega^{p,k}) = \Omega^{p,k}_{\lceil \frac{w}{2} \rceil} \tag{2.22}$$

$$\Omega^{p,k} = \left\{ |H(f)_k|^{p-\lfloor \frac{w}{2} \rfloor}, \ldots, |H(f)_k|^{p+\lfloor \frac{w}{2} \rfloor} : |H(f)_k|^{p-\lfloor \frac{w}{2} \rfloor} < |H(f)_k|^{p+\lfloor \frac{w}{2} \rfloor} \right\} \tag{2.23}$$

where $\Omega_{p,k}$ is the set containing w neighboring packet amplitudes $|H(f)_k|$ of the k-th subcarrier, arranged in ascending order.

The local MAD used to identify abnormal amplitude values is then defined as:

$$\sigma(\Omega^{p,k}) = \mu\left(|\Omega_i^{p,k} - \mu(\Omega^{p,k})|\right), \quad \forall i \text{ s.t. } 1 \leq i \leq w. \tag{2.24}$$

Finally, the acceptable local value intervals are defined as:

$$\text{limit}^{p,k} = \mu(\Omega^{p,k}) \pm \xi \cdot \sigma(\Omega^{p,k}), \tag{2.25}$$

and any value falling outside these ranges is replaced with the previous non-outlier value.

2.7 Phase

Another crucial signal characteristic derivable from Channel State Information (CSI) measurements is the signal phase. This measure pertains to the relationship between two or more radio signals sharing a common frequency at a given reference time t. In essence, multiple waves of identical frequency are deemed "in phase" if they are perfectly aligned. Conversely, if the crests and troughs of these waves do not overlap precisely, the radio waves are considered "out of phase."

The phase can be quantified as the relative displacement among these co-frequency radio signals, usually expressed in either degrees or radians. Figure 2.6 illustrates an original signal and its time-shifted counterpart, introducing a phase offset β.

It's pivotal to recognize that the phase can significantly impact the amplitude of the received radio wave. When signals are in phase, the combined amplitudes amplify the received signal strength. In contrast, when the signals are 180° out of phase, the resultant signal strength becomes nullified. This occurs because the crests of one wave align with the troughs of another wave, leading to cancellation.

Analogous to amplitudes, phases can be extracted from the CSI matrix. However, they necessitate additional processing to be harnessed effectively for Wi-Fi sensing applications Qian, Wu, Yang, Liu and Zhou (2014).

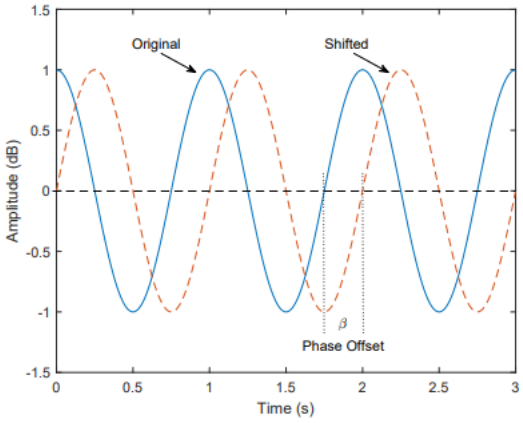


Figure 2.6 Example of a signal that is out of phase. The blue signal represents the original transmitted signal, and the dashed signal represents a delayed version of the same signal.

2.7.1 Phase Sanitization

The inherent randomness in signal phase, stemming from factors like random noise, unsynchronized clocks between transmitting and receiving antennas, or the Doppler effect due to relative motion between communicating devices, renders raw phase information unreliable. This randomness often leads to erratic phase offsets.

To rectify this issue, it is crucial to calibrate the CSI-extracted phases using a linear transformation-based technique, as advocated in the work of Sen, Radunovic, Choudhury and Minka (2012).

The raw CSI phase $\angle H_b(f)_\kappa$ measured for the κ-th subcarrier can be mathematically expressed as:

$$\angle \hat{H}(f)_\kappa = \angle H(f)_\kappa + 2\pi \frac{m_\kappa}{N} \Delta t + \beta + Z, \tag{2.26}$$

where $\angle H(f)_\kappa$ represents the real phase, Δt is the timing offset at the receiver, β is an unknown phase offset, Z is the measurement noise, m_κ and N correspond to the subcarrier index and Fast Fourier Transform (FFT) size, respectively, as specified by the IEEE 802.11n standard Halperin, Hu, Sheth and Wetherall (2010).

Since Δt and β are unknown, extracting genuine phase information directly is not feasible. However, by considering the phase across the entire frequency band, these unknown terms can be eliminated through a linear transformation, thereby mitigating the impact of random noise.

The slope a and offset b of the phase over the total frequency band can be mathematically represented as:

$$a = \frac{\angle \hat{H}(f)_K - \angle \hat{H}(f)_1}{m_K - m_1}, \tag{2.27}$$

$$b = \frac{1}{K} \sum_{k=1}^{K} \angle \hat{H}(f)_\kappa. \tag{2.28}$$

Finally, the calibrated phases $\angle \tilde{H}(f)_\kappa$ for the κ-th subcarrier are computed as:

$$\angle \tilde{H}(f)_\kappa = \angle H_b(f)_\kappa - a m_\kappa - b. \tag{2.29}$$

By subtracting the linear term $a m_\kappa + b$ from the raw measurement $\angle \hat{H}(f)_\kappa$, a linear combination of genuine phases, denoted as $\angle \tilde{H}(f)_\kappa$, is obtained, effectively eliminating the random phase offsets.

Chapter 3

Methodology

This chapter explores the central aspects of the project, outlining the methodologies and concepts that underpin it. I will discuss the dataset in detail, including its intricacies and the preprocessing techniques used to refine it. Additionally, I will analyze the architecture that was carefully designed and implemented to address the complex task of identifying materials. By thoroughly examining these components, my aim is to provide a clear understanding of the strategies employed and their underlying rationale.

3.1 Dataset

Since the task of material identification, is a relatively unexplored field, a dataset specifically created for this task was utilized. The dataset was created using measurements obtained through the ESP32 CSI toolkit (Hernandez and Bulut (2020)), which enables access to Channel State Information (CSI) and online processing from various devices. To obtain samples for the dataset, small cubes with the same dimensions but made of different materials were used (Figure 3.1). Specifically, the cubes consist of 13 different materials: acrylic, aluminum, brass, copper, empty, lignum vitae, nylon, oak, pine, polypropylene (pp), polyvinyl chloride (PVC), rosewood, and steel.

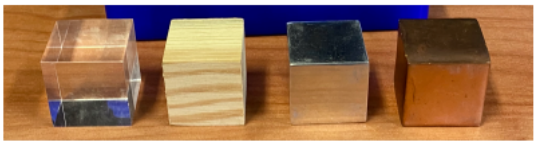


Figure 3.1 Some of the cubes used to collect CSI information. From left: Acrylic, Pine, Aluminium, and Copper

The measurements were conducted in two different ways: a shielded environment and an unshielded environment. In the first case, the measurements were taken in a shielded box to prevent any external interference (Figure 3.2a). In the second case, the box was opened to simulate a real environment with the possibility of external interference from other devices (Figure 3.2b). The box was coated with EMF shielding fabric to provide effective electromagnetic field protection. The various materials were exposed to Wi-Fi for different durations, and for each material, 10

different measurements were taken in both shielded and unshielded environments, resulting in a total of 260 measurements.



(a) Closed RF shield box.



(b) Open RF shield box.



(c) Final Setup.

Figure 3.2 Box used to collect materials CSI information

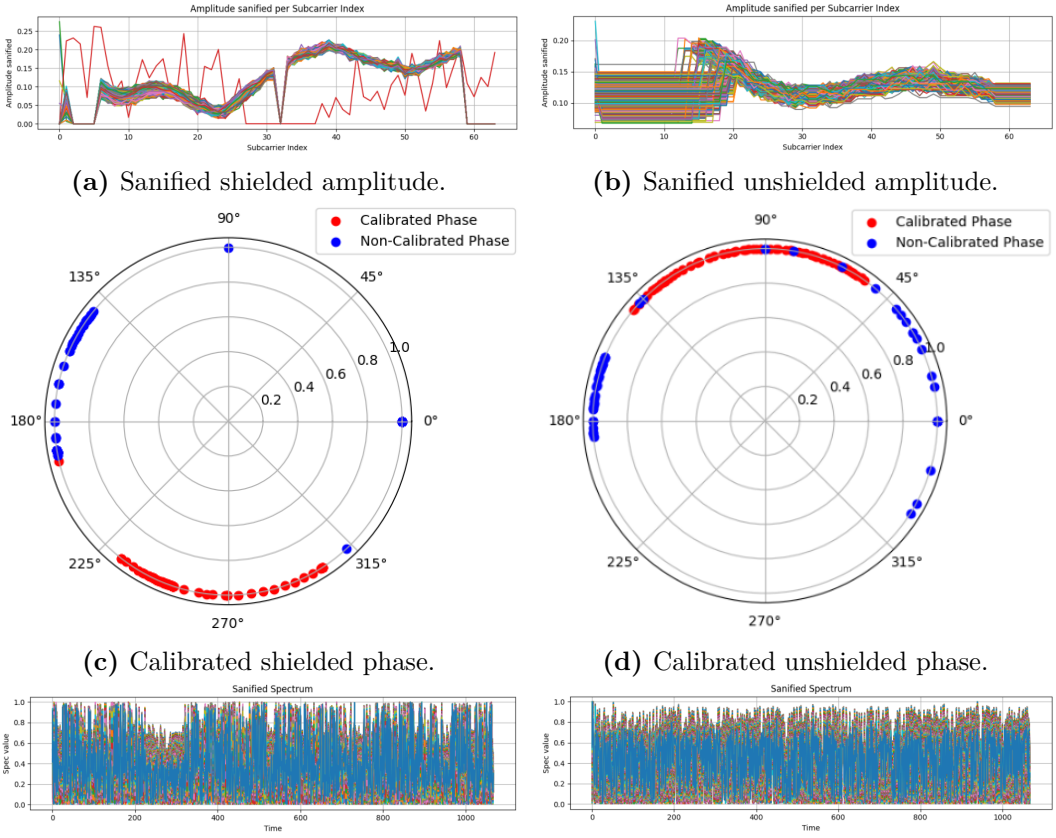
3.2 Preprocessing

For the preprocessing phase, both RSSI and CSI are required. Specifically, from the CSI, both amplitude and phase are extracted. As described in section 2.6 and section 2.7, the raw values of amplitude, phase, and CSI in general, cannot be considered as they are very noisy. Therefore, before being passed to the network, the amplitude and phase are sanitized using the outlier removal and calibration methods described in (subsection 2.6.1, subsection 2.7.1). Another crucial element is the spectrogram. Inspired by the uses of CSI spectrograms, particularly in fall detection applications (Nakamura et al. (2020)), the spectrogram was obtained by analyzing all samples of the materials to identify the time instants exposed to Wi-Fi electromagnetic waves. The first 1068 time instants were specifically chosen to extract the spectrogram. This decision was made because the material with the fewest measurements had exactly 1068 elements in terms of time instants. The spectrogram is obtained by applying the short-time Fourier transform (STFT) to the CSI series. However, a sanitization operation is necessary as the extracted spectrogram would be too noisy otherwise. For this reason, the Hann window is used. Specifically, a window of length 91 is used. Subsequently, the spectrogram is normalized and then interpolated to have a size of 224x224. This allows the spectrogram to be processed by the ResNet network (He, Zhang, Ren and Sun (2015)). The result of pre-processing applied on phase amplitude and spectrogram is shown in Figure 3.3.

In addition, a complete workflow of the process is illustrated in Figure 3.4

3.3 Training process

The training phase starts with an in-depth exploration of the dataset composition, comprising 10 tests for each class with varying CSI measurements across tests. To ensure robust experimentation, the dataset was partitioned into distinct subsets: 80% for the training set, 10% for the validation set, and 10% for the test set. These partition percentages were chosen strategically due to the relatively limited number of test samples available in the dataset.



(a) Sanified shielded amplitude.

(b) Sanified unshielded amplitude.

(c) Calibrated shielded phase.

(d) Calibrated unshielded phase.

(e) Sanified shielded spectrum.

(f) Sanified unshielded spectrum.

Figure 3.3 Examples of signal sanitization for shielded and unshielded data, respectively

A critical challenge encountered during training was the risk of overfitting, a common issue in neural networks where the model memorizes the training data but fails to generalize to unseen data. This can occur for several reasons, such as using complex models for simple tasks, the presence of noisy data, or insufficient data. To prevent this issue, a detailed analysis of the dataset was conducted. It was observed that some classes had more samples per file due to varying WiFi exposure times for different materials, leading to an imbalance in sample numbers. Consequently, a class balancing strategy was implemented, and the model weights were initialized to give more importance to classes with fewer total samples. The total number of samples for each class can be seen in Table 3.1.

The class balancing function operates by extracting the frequencies of each class from both shielded and non-shielded datasets. The reciprocal of each frequency is then calculated, assigning greater weight to less frequent classes. These frequencies are then normalized so that the sum of the weights equals the number of classes in the dataset.

Moreover, to mitigate overfitting further, an early stopping mechanism was integrated into the training process. This algorithm counts the number of epochs where the current validation loss exceeds the validation loss obtained at least half of the time within a window of eight epochs. If this count reaches five, the training is

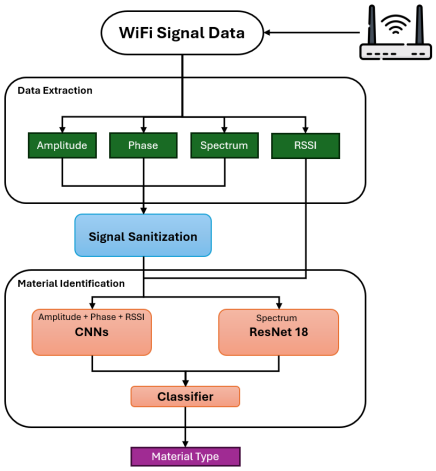


Figure 3.4 The figure illustrates the workflow of the entire process.

Class	Shielded Dataset samples	Unshielded Dataset samples	Total samples
Acrylic	25358	30223	55581
Aluminium	37560	33536	71096
Brass	32032	31105	63137
Copper	11661	15954	27615
Empty	18464	30293	48757
Lignum vitae	29420	21880	51300
Nylon	44603	48489	93092
Oak	35503	22373	57876
Pine	14071	16031	30102
PP	25366	21152	46518
PVC	31239	33614	64853
Rosewood	28233	36624	64857
Steel	40734	31846	72580

Table 3.1 Total number of measurements for each material.

stopped. This multi-faceted approach aimed to enhance the model's generalization capabilities and mitigate the risk of overfitting during training.

3.4 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a specialized type of network that excels particularly when dealing with images or signals. At the core of CNNs are three types of layers: Convolutional layers, pooling layers, and Fully-connected (FC) layers. In particular, a CNN is composed of one or more convolutional layers or pooling layers and finally fully connected layers, which are the final layers of the network.

Analyzing the various layers of this type of network, we can say that the convolutional layer represents the core building block of a CNN. It is where the majority of computations take place and requires three main components: input data, filters (also known as feature detectors), and a feature map.

If we assume an image as input data, then we would be dealing with a 3D input

(height, width, and depth). Additionally, we would also have a feature detector, also known as a kernel or filter, which will move across the receptive fields of the image, checking if the feature is present. This process is known as convolution.

The feature detector is typically a 2D array of weights that scans receptive fields in the image. Most of the time, the dimension of the kernel is a 3x3 matrix. The process is illustrated in Figure 3.5.

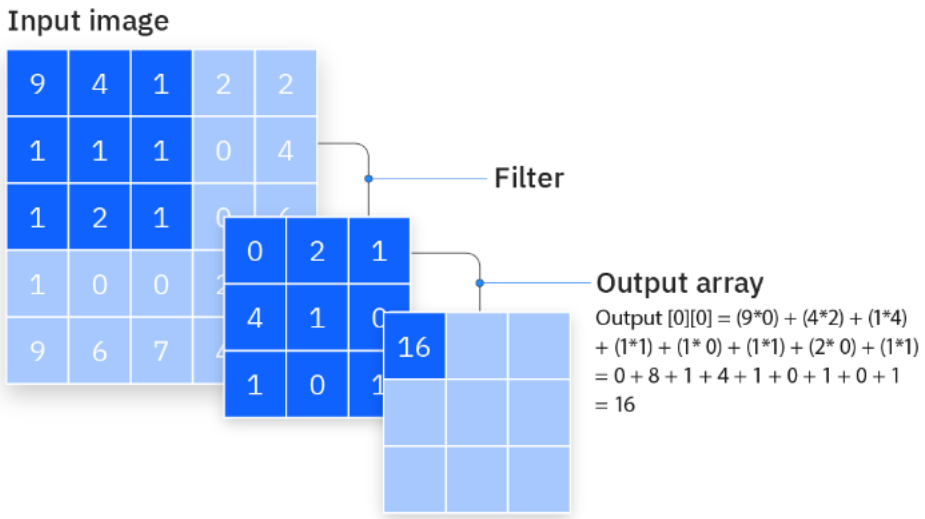


Figure 3.5 Example of convolution operation in the convolutional layer.

After each convolution operation, a CNN applies a Rectified Linear Unit (ReLU) transformation to the feature map, introducing nonlinearity to the model. Pooling layers, on the other hand, serve to perform dimensionality reduction, allowing for a reduction in the number of parameters in the input. Its operation is very similar to that of the kernel; it sweeps a filter across the entire input, but the pooling layer does not have weights. There are two types of pooling: max pooling, which selects the pixel with the maximum value to send to the output array, and average pooling, which calculates the average value within the receptive field to send to the output array.

Finally, we have the fully-connected layer. In previous cases, the pixel values of the input image are not directly connected to the output layer in partially connected layers. On the contrary, in the fully-connected layer, each node in the output layer connects directly to a node in the previous layer.

These layers are used to perform the task of classification based on the features extracted through the previous layers and their different filters. While convolutional and pooling layers tend to use ReLU functions, FC layers usually leverage a softmax activation function to classify inputs appropriately, producing a probability from 0 to 1.

3.5 Residual Neural Network

Residual Neural Networks (ResNets) are deep learning models primarily used in computer vision. Their architecture is similar to that of CNNs but modified to support thousands of convolutional layers. CNNs were not able to scale to a large number of layers, which resulted in limited performance. Additionally, adding more layers led researchers to encounter the so-called vanishing gradient problem. Neural networks are trained through the process of backpropagation, which relies on gradient descent, shifting down the loss function and finding the weights that minimize it. If there are too many layers, repeated multiplications will eventually reduce the gradient until it "disappears," and performance saturates or deteriorates with each layer added. ResNet was developed to provide an innovative solution to the vanishing gradient problem, known as "skip connections."

ResNet stacks multiple identity mappings (convolutional layers that do nothing at first), skips those layers, and reuses the activations of the previous layer. Skipping speeds up initial training by compressing the network into fewer layers. Then, when the network is retrained, all layers are expanded, and the remaining parts of the network, known as the residual parts, are allowed to explore more of the feature space of the input image.

One of the key concepts of ResNet is the residual blocks. In old architectures like VGG16 (Simonyan and Zisserman (2015)), convolutional layers are stacked with batch normalization and nonlinear activation layers such as ReLU between them. The ResNet architecture introduces the simple concept of adding an intermediate input to the output of a series of convolution blocks.

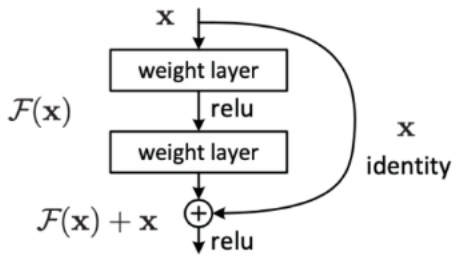


Figure 3.6 Residual learning: a building block.

This technique smooths out the gradient flow during backpropagation, enabling the network to scale to 50, 100, or even 150 layers. Skipping a connection does not add additional computational load to the network.

3.6 Architecture

The architecture comprises two Convolutional Neural Networks (CNNs), a 2D and a 1D, alongside a ResNet18. These neural networks were selected for their efficacy in analyzing radio signals, with CNNs excelling in signal processing tasks and ResNet commonly applied in image or spectrogram-related tasks (Nakamura et al. (2020)).

The 2D CNN is designed to process amplitude, phase, and RSSI. It comprises three convolutional layers with dimensions of 64, 128, and 256. Following each

convolutional layer, a leaky ReLU activation function is applied, along with dropout regularization.

The 1D CNN, instead, focuses on processing the phase. It mirrors the 2D CNN in terms of layer dimensions (64, 128, 256), but employs a kernel size of 3 and a stride of 2. Similar to the 2D CNN, each convolutional layer is followed by the application of a leaky ReLU and dropout. Subsequently, the outputs of both CNNs are concatenated and fed into two linear layers with leaky ReLU activation, followed by a linear classifier.

The ResNet network, assigned to process the spectrogram, has been modified by replacing the last layer, enabling it to produce an output of size 13, corresponding to the number of classes. Finally, the combined outputs of the ResNet and the CNNs are concatenated and further processed by a linear classifier. The architectural layout is illustrated in Figure 3.7. Notably, MatID's training was conducted using both shielded and unshielded datasets to ensure robustness and adaptability across different scenarios.

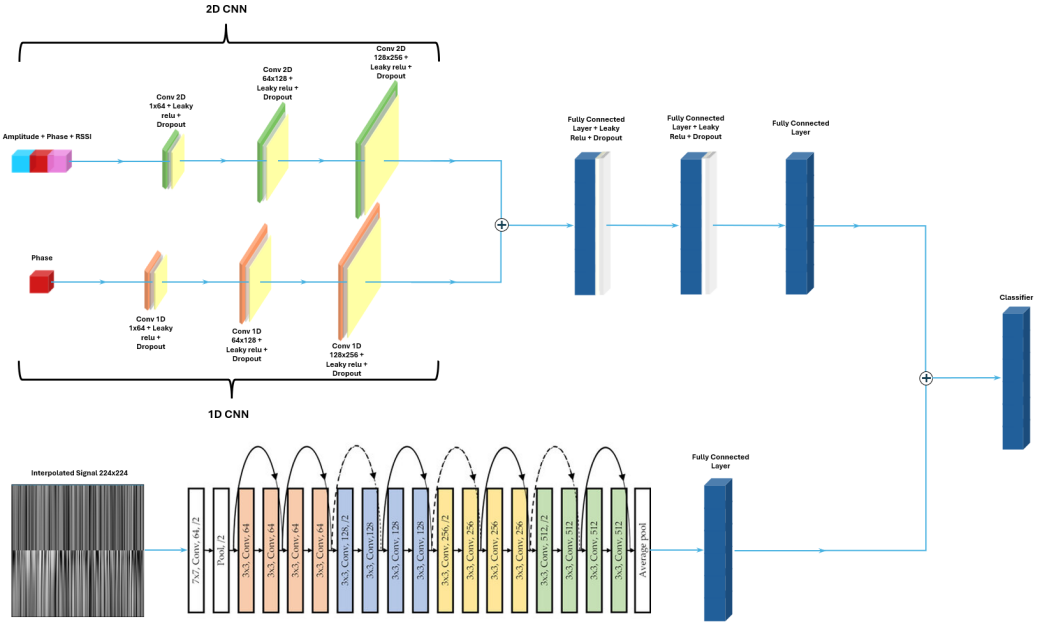
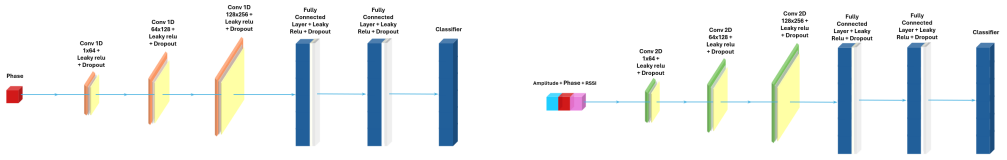


Figure 3.7 Complete architecture of MatID.

Chapter 4

Experiments and Results

This chapter focuses on the practical implementations proposed for the study of material identification. I will conduct a thorough analysis of the outcomes derived from these implementations, elucidate the training methodologies employed, and discuss the different strategies deployed to enhance the model's performance.



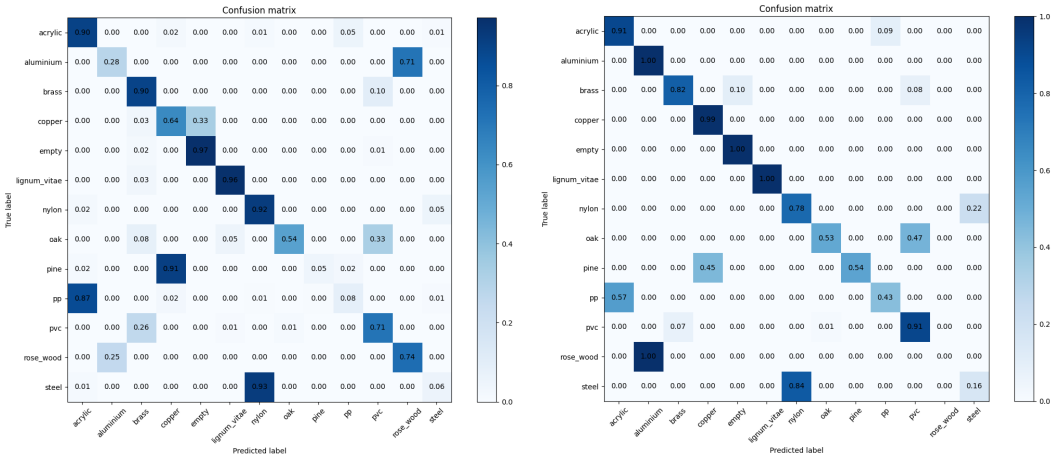
(a) Model Architecture for the 1D CNN. **(b)** Model Architecture for the 2D CNN.

Figure 4.1 Previous model architectures.

4.1 Results with previous models

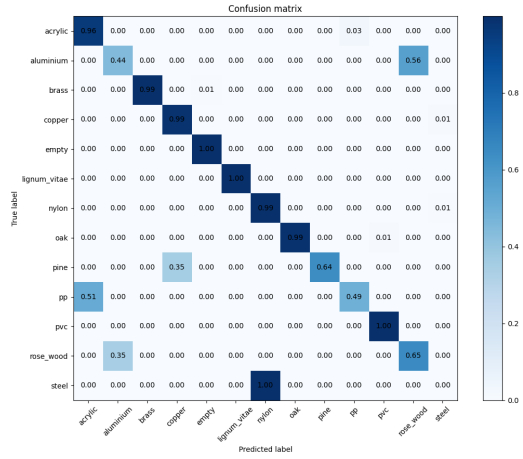
Various experiments were conducted using different models, including a 2D CNN and a 1D CNN. The architectures of these models are depicted in Figure 4.1. Both models were tested combined and individually using amplitude and phase values, and their results were analyzed. The results are shown in Figure 4.2 and Figure 4.3. Furthermore, tables displaying comparative results can be found in Table 4.3 and Table 4.4.

The most evident aspect that emerges from the following results is that despite the combination of phase and amplitude producing very similar results for both architectures, analyzing them individually revealed distinct trends. Specifically, the 2D CNN exhibited poor performance when processing the single phase (Figure 4.3a, Figure 4.3d), unlike the 1D CNN, which yields much more significant results (Figure 4.2d, Figure 4.2a). The same applies to the amplitude. It seems that the 2D CNN produces better results compared to the 1D CNN. For this reason, it was decided to use the information from the 1D CNN to process the phase and the 2D CNN to process the amplitude. This result was certainly unexpected since temporal signals such as amplitude and phase should be processed more efficiently by CNN1Ds

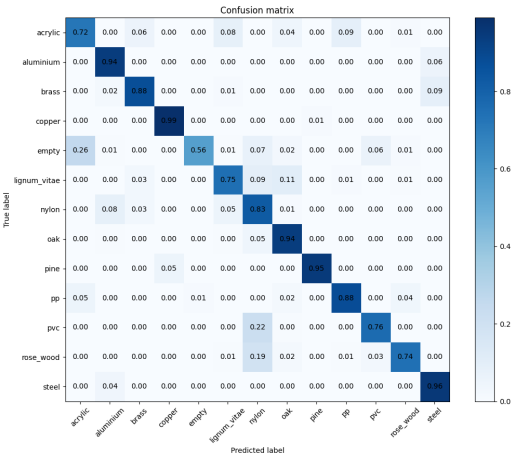


(a) Unshielded phase.

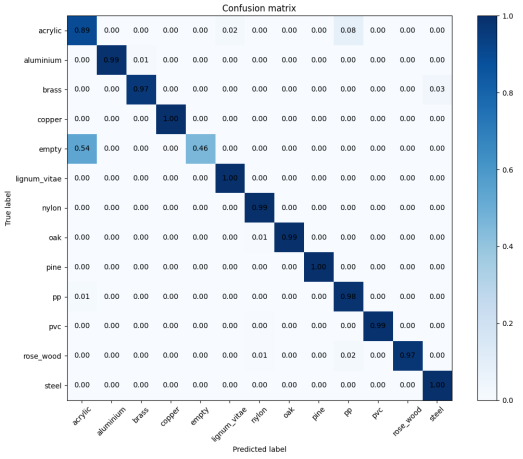
(b) Unshielded amplitude.



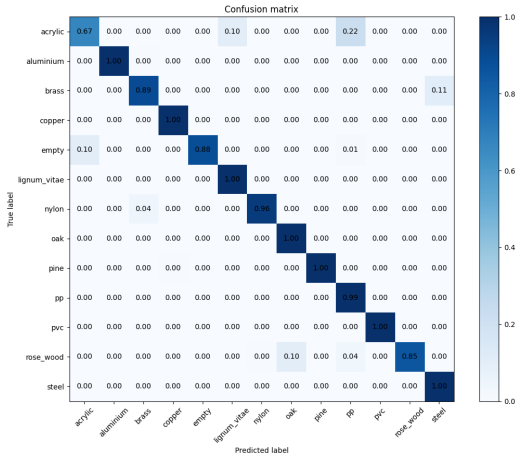
(c) Unshielded amplitude and phase.



(d) Shielded phase.

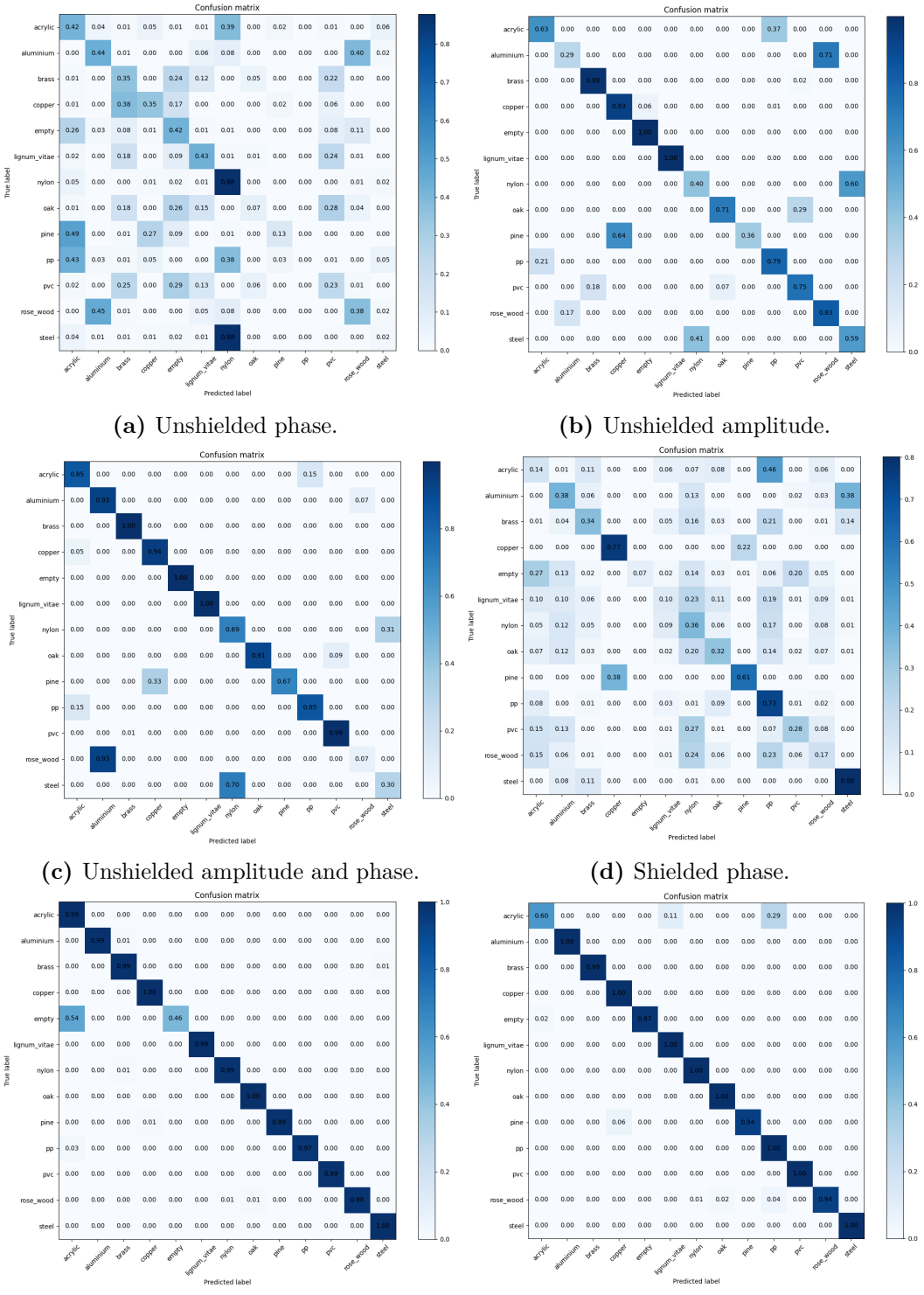


(e) Shielded amplitude.



(f) Shielded amplitude and phase.

Figure 4.2 Results for unshielded and shielded Amplitude, Phase and Amplitude + Phase processed by CNN 1D.



(a) Unshielded phase.

(b) Unshielded amplitude.

(c) Unshielded amplitude and phase.

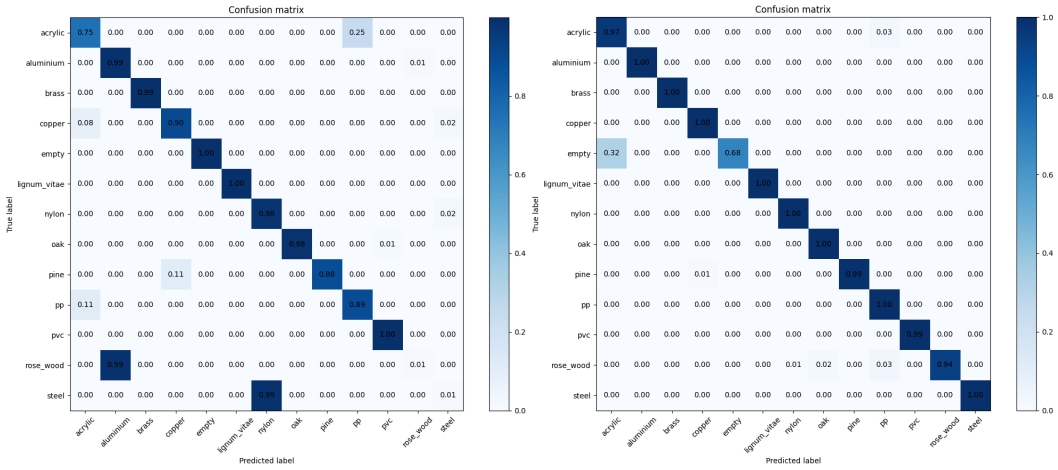
(d) Shielded phase.

(e) Shielded amplitude.

(f) Shielded ampltude and phase.

Figure 4.3 Results for unshielded and shielded Amplitude, Phase and Amplitude + Phase processed by CNN 2D.

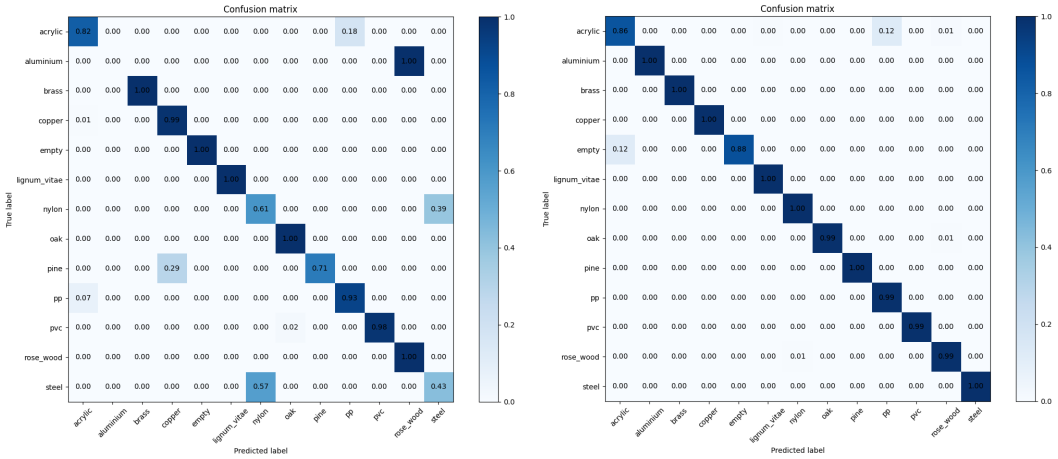
but this happens only for one of the signals in question. However, the combination of these two architectures improved the results, but there could have been further room for improvement, especially regarding the tests conducted with unshielded data.



(a) Unshielded amplitude + phase + RSSI. **(b)** Shielded amplitude + phase + RSSI.

Figure 4.4 Results for unshielded and shielded Amplitude + Phase + RSSI processed by CNN 2D.

In this regard, it was exploited the fact that the combination of RSSI, amplitude, and phase, processed with the 2D CNN, obtained satisfactory results as can be seen in Table 4.3 and Table 4.4and Figure 4.4. For this reason, it was decided to let both the 1D CNN and the 2D CNN process the phase, along with amplitude and RSSI. The results obtained by applying this technique are shown in Figure 4.5.



(a) **(b)**

Figure 4.5 Results for unshielded and shielded Amplitude Phase and RSSI processed by the combination of CNN1D and CNN2D.

As can be seen, the results have improved compared to previous implementations,

but there are still some issues when dealing with tests on unshielded data.

4.2 Final Results

To further enhance the results, it was decided to utilize an element commonly used in other areas of WiFi sensing, namely, the spectrogram. Spectrograms were extracted from the CSI measurements of each test. These spectrograms, which were sanitized using the procedure outlined in (section 3.2), were analyzed using a ResNet18, thereby finalizing MatID's structure (Figure 3.7).

For the selection of the best hyperparameters, a grid search approach was used. The learning rate of the CNNs was chosen from {0.0001, 0.001, 0.01, 0.1}, as was the learning rate of the ResNet {0.0001, 0.001, 0.01, 0.1}. Weight decay was also used, selecting it for both the CNNs and the ResNet from the following values {0.0001, 0.001, 0.01, 0.1}. The dropout value was selected from the following values {0.2, 0.5, 0.8}. A batch size of 64 was selected to balance performance and speed. Adam (Adaptive Moment Estimation) was chosen as the optimizer because it is one of the best optimization algorithms currently available. Indeed, Adam works with momentums of the first and second order. The intuition behind Adam is that we don't want to roll so fast just because we can jump over the minimum, we want to decrease the velocity a bit for a careful search. All of this allows Adam to be fast and converge quickly, and it also rectifies vanishing learning rates and high variance.

The optimal set of hyperparameters is shown in Table 4.1.

Hyperparameters	
Learning Rate CNNs	0.0001
Learning Rate ResNet	0.0001
Weight Decay CNNs	0.0001
Weigh Decay ResNet	0.0001
Dropout	0.5
Batch Size	64
Optimizer	Adam

Table 4.1 Optimal hyperparameters set.

The results in terms of F1-score, accuracy, precision, and recall are shown in Table 4.2. Figure 4.6 is the confusion matrix obtained with MatID. As can be seen, the network successfully classifies most of the classes except for a few, possibly because in these cases, the CSI measurements are too similar to those of other materials.

	Score Shielded + Unshielded	**Score Shielded**	**Score Unshielded**
F1	0.9863	0.9890	0.9830
Accuracy	0.9863	0.9890	0.9830
Precision	0.9867	0.9896	0.9840
Recall	0.9863	0.9890	0.9830

Table 4.2 MatID's score.

Accuracy represents the number of correctly classified data instances over the total number of data instances. However, accuracy may not be a good measure if the dataset is not balanced. In the case of this study, the classes are not perfectly balanced in terms of measurements; therefore, other measurements have been taken into consideration. Precision (also called positive predictive value) is the fraction of relevant instances among the retrieved instances. Recall (also known as sensitivity) is the fraction of relevant instances that were retrieved. However, one of the most effective measurements is the F1-score, which represents the harmonic mean of precision and recall. Commonly used as an evaluation metric in binary and multi-class classification and LLM evaluation, the F1 score integrates precision and recall into a single metric to gain a better understanding of model performance.

	Shielded Data			
Model	**Amplitude**	**Phase**	**Amplitude + Phase + RSSI**	**Amplitude + Phase + RSSI + Spectrum**
CNN 1D	0.9518	0.8455	0.9460	/
CNN 2D	0.9576	0.3994	0.9719	/
CNN 1D + CNN 2D	/	/	0.9820	/
MatID	/	/	/	0.9890

Table 4.3 Scores of previous models and MatID with shielded data.

4.3 Comparative Results

In this section, we compare the performance of the proposed MatID model with other state-of-the-art (SOTA) models for material identification discussed in section 1.1. Although the specific methodology and datasets used in this thesis differ from those of the related works, this comparison aims to demonstrate the effectiveness of the proposed approach.

The performance of MatID is evaluated in terms of accuracy and compared to the results obtained by other SOTA models. As detailed in the Related Works section, these models have set a benchmark in the field of material identification. The comparative analysis highlights the strengths of MatID, showcasing its competitive performance despite the differing experimental setups. The comparative results are summarized in Table 4.5.

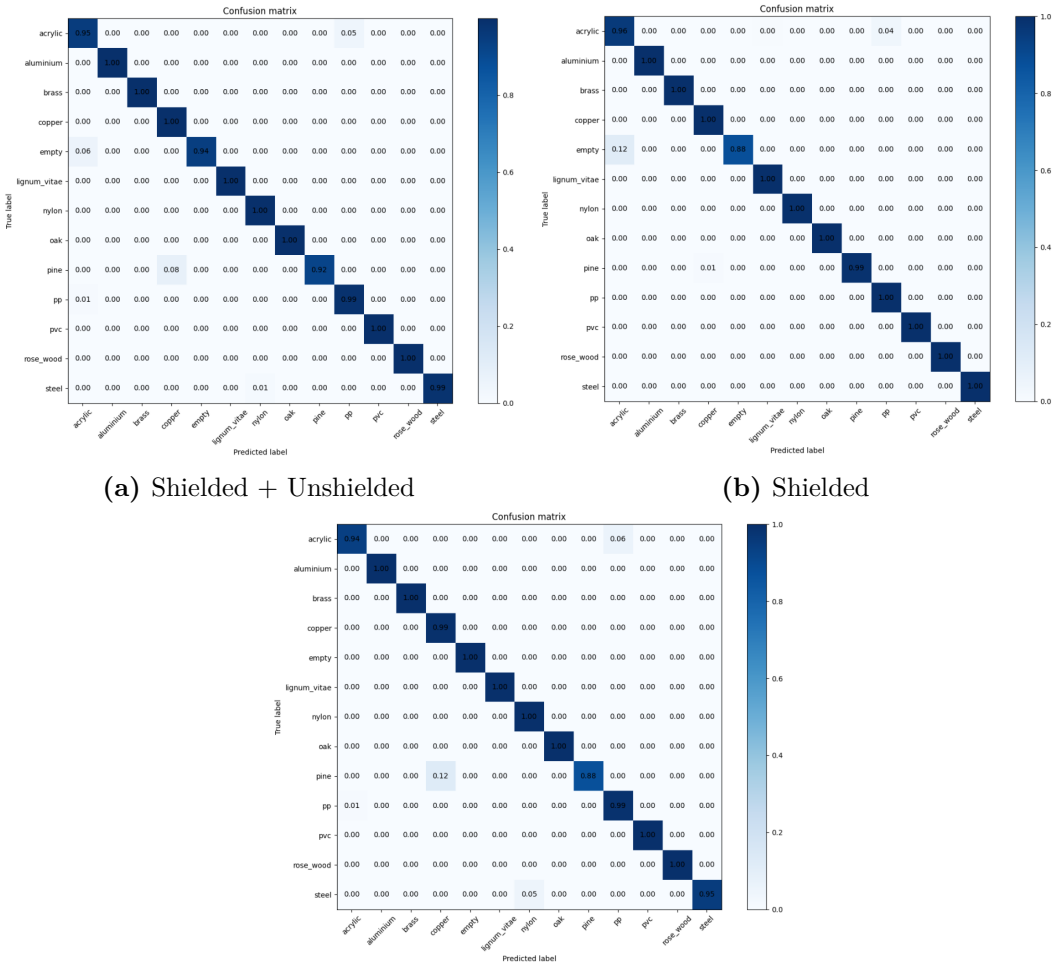
The results indicate that MatID not only meets but, in some cases, exceeds the performance of existing models, underscoring the robustness and reliability of the proposed method in material identification tasks.

	Unshielded Data			
Model	**Amplitude**	**Phase**	**Amplitude + Phase + RSSI**	**Amplitude + Phase + RSSI + Spectrum**
CNN 1D	0.6582	0.5435	0.7590	/
CNN 2D	0.7062	0.2784	0.7882	/
CNN 1D + CNN 2D	/	/	0.7978	/
MatID	/	/	/	0.9830

Table 4.4 Scores of previous models and MatID with unshielded data.

Models	Accuracy
WiMi (Feng et al. (2019))	96%
Shi et al. (2021)	97%
WiMate (Gu et al. (2021))	96.2%
MatID	98.63%

Table 4.5 Comparative results with other SOTA models.



(a) Shielded + Unshielded

(b) Shielded

(c) Unshielded

Figure 4.6 MatID results with shielded + unshielded, shielded and unshielded dataset.

Chapter 5

Conclusions

This thesis successfully demonstrates the potential of WiFi Sensing technology in non-intrusive material identification. By using advanced preprocessing techniques and deep learning models, the study addresses key challenges such as signal noise, phase distortion, and spectrogram noise, which enhances the accuracy and robustness of material classification. Advanced signal processing plays a significant role in this study. The introduction of rigorous preprocessing routines, including amplitude sanitization, phase calibration, and spectrogram sanitization, greatly improves the quality of the signal data and the reliability of the input data used for neural network processing. The integration of Convolutional Neural Networks (CNNs) and Residual Neural Networks (ResNet) allows for effective extraction and analysis of signal features. The combination of 1D and 2D CNNs with ResNet18 proves particularly effective in handling the complex data characteristics of WiFi signals. The robust experimental setup, which includes the use of both shielded and unshielded environments, provides a comprehensive evaluation of the models' performance, ensuring their applicability in real-world scenarios.

The research implications of these findings highlight the feasibility of using WiFi signals for detailed environmental sensing and object detection, presenting a cost-effective and privacy-preserving alternative to traditional visual methods. This technology has promising applications in various fields, including security, healthcare, and smart environments.

Looking towards future directions, enhanced noise reduction could be a focus. Future work could explore more sophisticated noise reduction techniques to further refine signal quality. Additionally, expanding the dataset to include a wider range of materials and conditions would enhance the model's generalizability. Developing real-time processing capabilities could pave the way for practical implementations in dynamic environments. Overall, this thesis lays a foundation for the advancement of WiFi Sensing technology, demonstrating its viability and potential for broad applications in non-invasive sensing and material identification.

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